

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning,*

and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4

Total

Marks:40

Annexure A

ARTICLE REVIEW

Code of Conduct:

I ____BAIRAGANI DINESH -192372189____ (Name / Reg No) certify that this submission is my original work and that I have adhered

to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: BAIRAGANII DINESH-192372189

Annexure A

ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications.*

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

(a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

(b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

(a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

(b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.

(b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?

(c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

(a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.

(b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article’s purpose,	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article

		concepts, and arguments			
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well- structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganize d and difficult to understand

ARTICLE REVIEW

Article Selected

Title: Efficient Detection of Botnet Traffic by Features Selection and Decision Trees

Authors: Javier Velasco-Mata, Víctor González-Castro, Eduardo Fidalgo Fernández, and Enrique Alegre

Journal: IEEE Access, Volume 9, pp. 120567–120579

Year: 2021

DOI: 10.1109/ACCESS.2021.3108222

Domain/Application Area: Cybersecurity and Network Traffic Classification

ML Problem: Classification of network traffic into botnet-related classes using machine learning and decision-tree-based models.

Article Link: <https://doi.org/10.1109/ACCESS.2021.3108222>

The article was published in IEEE Access in 2021, satisfying the requirement that the selected research article should be published after 2019. ([ResearchGate](#))

TASK 1: ARTICLE SUMMARY

(a) Citation Details and Problem Description

Velasco-Mata, J., González-Castro, V., Fidalgo Fernández, E., & Alegre, E. (2021).

“Efficient Detection of Botnet Traffic by Features Selection and Decision Trees.”
IEEE Access, 9, 120567–120579. DOI: 10.1109/ACCESS.2021.3108222. ([DBLP](#))

The article focuses on **cybersecurity**, particularly the detection and classification of botnet traffic. Botnets are networks of compromised devices controlled by attackers and can be used for malicious activities such as DDoS attacks and spam campaigns. The main machine-learning problem addressed is the efficient classification of network traffic into different botnet classes.

(b) Key Objective and Research Gap

The main objective of the research is to develop an efficient machine-learning-based botnet traffic detection system that can maintain good classification performance while reducing the number of network traffic features and classification time. Traditional network security systems often depend on signatures, but these approaches may become ineffective when attackers modify or encrypt their communications. Therefore, behavior-based machine learning methods can provide a more flexible alternative.

The authors identified a research gap in the **efficiency of trained botnet detectors**, especially the time required to classify network traffic after the model has been trained. Previous research mainly concentrated on improving detection accuracy or F1-score, while less attention was given to reducing the number of features and improving classification speed.

To address this gap, the authors applied two feature-selection techniques: **Information Gain and Gini Importance**. These methods were used to create feature subsets containing five, six, and seven features. The selected features were then evaluated using **Decision Tree (DT)**, **Random Forest (RF)**, and **k-Nearest Neighbors (k-NN)** models. Two datasets, QB-CTU13 and EQB-CTU13, were generated from the CTU-13 dataset. ([ResearchGate](#))

TASK 2: METHODOLOGY ANALYSIS

(a) Machine Learning Techniques and Dataset

Machine Learning Techniques

The study uses three supervised machine learning algorithms:

1. Decision Tree (DT)

- A tree-based supervised learning algorithm.
- It recursively divides the data according to important features.
- It is easy to interpret and relatively fast during prediction.

2. Random Forest (RF)

- An ensemble method consisting of multiple decision trees.
- The predictions of several trees are combined to improve generalization.
- It can reduce some of the overfitting problems associated with a single decision tree.

3. k-Nearest Neighbors (k-NN)

- A distance-based classification method.
- It classifies a new sample according to the classes of its nearest training samples.

Feature Selection

Two feature-selection methods were used:

Information Gain:

Measures how much information a feature provides about the target class.

Gini Importance:

Measures the importance of features based on their contribution to reducing impurity in tree-based models.

These techniques produced feature subsets containing **five, six, and seven features**, allowing the researchers to investigate whether fewer features could provide efficient botnet classification. ([Directory of Open Access Journals](#))

Dataset

The study is based on the **CTU-13 botnet dataset**. The authors generated two modified datasets:

- **QB-CTU13**
- **EQB-CTU13**

The purpose of creating these datasets was to address class imbalance and provide suitable data for evaluating botnet classification models. The datasets contain network-traffic-related features and multiple botnet classes.

The authors selected reduced feature subsets rather than using the complete feature set. This reduction helped them study both classification performance and prediction speed. ([Bulería](#))

(b) Mapping to CO4

CO4 Topic	Connection with Article
Inductive Learning	The models learn patterns from labelled network traffic examples and generalize them to unseen traffic.
Decision Trees	Decision Tree is one of the main classification algorithms evaluated.
Statistical Learning	Feature selection and supervised classification are used to learn relationships between traffic features and botnet classes.
Reinforcement Learning	Not directly used in this article.

Methodological Strength

A major strength is the combination of **feature selection and classification**. Instead of concentrating only on classification accuracy, the authors also investigate the computational time needed to classify traffic. This makes the approach more suitable for practical cybersecurity systems where fast detection is important.

Methodological Limitation

One limitation is that the experiments depend mainly on datasets derived from **CTU-13**. Real-world network traffic can be more diverse and can contain new or previously

unseen botnet behaviors. Therefore, performance on CTU-13 may not completely represent performance in a real enterprise network.

TASK 3: RESULTS AND CRITICAL EVALUATION

(a) Key Results

The study evaluates Decision Tree, Random Forest, and k-NN using different feature subsets. The main performance measure is the **macro-averaged F1-score**, together with the computational time required to classify a sample.

The most important reported result is that the **Decision Tree using five selected features achieved a mean F1-score of approximately 85% with an average classification time of 0.78 microseconds per sample.** ([ResearchGate](#))

Comparative Summary

Model / Feature Strategy	Main Observation
Decision Tree + 5 features	Best overall efficiency; mean F1 $\approx$ 85%
Decision Tree + more features	Performance varies by botnet class
Random Forest	Provides ensemble-based classification but has greater computational complexity
k-NN	Provides a distance-based comparison but can require more computation during prediction
Feature Selection	Reduced the feature space and improved efficiency

The authors also report that the five-feature Decision Tree achieved particularly strong results for some individual botnet classes. For example, recall values of approximately **99% for Miuref** and **97% for NotPetya** were reported in the detailed evaluation. However, the Bunitu class remained more challenging, with an F1-score of approximately **84% using the optimized five-feature Decision Tree.** ([ResearchGate](#))

(b) Critical Evaluation

The reported results are useful because the authors evaluate both **classification quality and prediction speed**. An F1-score of around 85% is reasonable for a multi-class cybersecurity problem because botnet traffic can be difficult to distinguish, especially when classes are imbalanced.

However, several issues should be considered.

Possible Biases and Limitations

1. Dataset bias:

The experiments are based on CTU-13-derived datasets. Network environments in real organizations may have different traffic patterns.

2. Class imbalance:

Botnet datasets can contain substantially different numbers of samples for different classes. Although the authors created QB-CTU13 and EQB-CTU13 to address this issue, imbalance can still influence classification results.

3. Generalization:

A model that performs well on known botnet classes may not detect a completely new botnet family effectively.

4. Limited feature representation:

Reducing the feature set improves efficiency, but some information useful for detecting new attacks may be lost.

5. Single-model overfitting:

The authors themselves recognize that a single Decision Tree can be susceptible to overfitting. ([ResearchGate](#))

Additional Experiments That Would Strengthen the Research

The research could be strengthened by:

- Testing the model on additional modern botnet datasets.
- Performing cross-dataset validation.
- Testing unseen or zero-day botnet families.
- Comparing the Decision Tree with modern ensemble models such as XGBoost.
- Performing repeated experiments with different train-test splits.
- Reporting confidence intervals for the performance metrics.
- Testing the system on live network traffic.
- Evaluating memory usage and CPU utilization in addition to classification time.

(c) Real-World Applicability

The proposed approach can be deployed in:

- Enterprise intrusion detection systems.
- Data-center network monitoring.

- Internet Service Provider networks.
- IoT security systems.
- University and campus networks.
- Security Operations Centers (SOCs).
- Real-time network traffic monitoring systems.

The main advantage is its low classification time and reduced feature requirement. A Decision Tree with a small number of important features can potentially be integrated into a real-time intrusion detection pipeline.

However, industrial deployment introduces several challenges.

Data Availability

Organizations need large quantities of labelled network traffic. Obtaining accurately labelled malicious traffic is difficult because security events are relatively rare.

Scalability

High-speed enterprise networks can generate millions of packets or flows. The detection system must process this traffic without becoming a bottleneck.

New Attack Types

Attackers continuously develop new botnets. A model trained on older botnet families may not perform well against previously unseen attacks.

Privacy and Ethical Issues

Network monitoring can involve sensitive information. Organizations must ensure that traffic monitoring follows privacy regulations and internal security policies.

Model Maintenance

The model should be periodically retrained with recent traffic because network behavior and attack techniques change over time.

TASK 4: REFLECTION AND LEARNING OUTCOME

(a) Three Key Insights

Insight 1: Importance of Feature Selection

I learned that using more features does not always produce the best machine-learning solution. Selecting the most informative features can reduce computational requirements while maintaining useful classification performance.

Insight 2: Decision Trees Can Provide Fast Classification

The article demonstrated that Decision Trees can be useful when fast prediction is important. The five-feature Decision Tree achieved a mean F1-score of about 85% while requiring only about 0.78 microseconds per sample on average. ([Directory of Open Access Journals](#))

Insight 3: Accuracy Alone Is Not Enough

Another important learning is that evaluating a machine-learning model only through accuracy is insufficient. In cybersecurity, the model should also be evaluated using F1-score, recall, classification time, computational requirements, and its ability to generalize.

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(b) Proposed Extension

I would extend this research by developing a **hybrid botnet detection system using XGBoost with adaptive feature selection and testing it on multiple recent network-security datasets.**

The proposed experiment would compare:

Decision Tree → Random Forest → XGBoost → k-NN

using the same selected feature sets. The models would be evaluated using:

- Accuracy
- Precision
- Recall
- Macro F1-score
- AUC-ROC
- Detection time
- Memory consumption

The experiment would also include cross-dataset testing, where the model is trained using one dataset and tested on another. This would provide a stronger indication of real-world generalization.

The enhancement is feasible because XGBoost and feature-selection methods are widely available in Python machine-learning libraries. The expected impact is improved detection of different botnet classes while maintaining reasonable prediction speed. It could also help determine whether a more advanced ensemble model provides a better balance between **accuracy, robustness, and computational efficiency** than a single Decision Tree.

CONCLUSION

The reviewed article presents an effective approach for improving botnet traffic detection using feature selection and Decision Tree-based machine learning. The researchers demonstrated that reducing the number of input features can provide efficient classification without requiring a large feature set. The Decision Tree using five selected features achieved a mean F1-score of approximately 85% and an average classification time of 0.78 microseconds per sample. ([ResearchGate](#))

The study is strongly connected with CO4 because it demonstrates **supervised inductive learning, Decision Tree classification, feature selection, and machine-learning-based decision-making**. Its main contribution is not simply achieving good classification performance, but showing that an efficient model can also be developed for fast botnet traffic classification.

Overall, the article helped me understand that a practical machine-learning system must consider not only predictive performance but also feature complexity, computational efficiency, generalization, and real-world deployment requirements.